

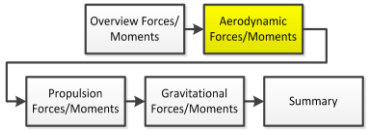
Introduction To Flight Mechanics & Control

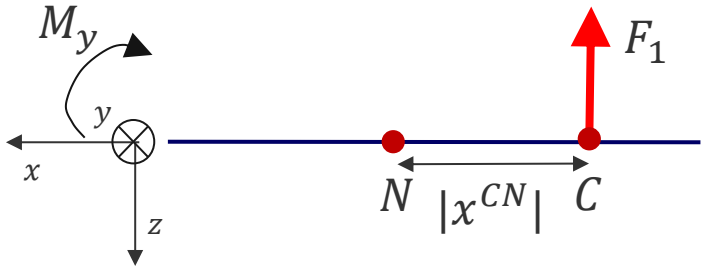
Exercise 3

Exercise 1: Neutral Point of the Wing

a) What holds true for the pitch moment around the neutral point?

b) How does the position of the center of pressure depend on the angle of attack?

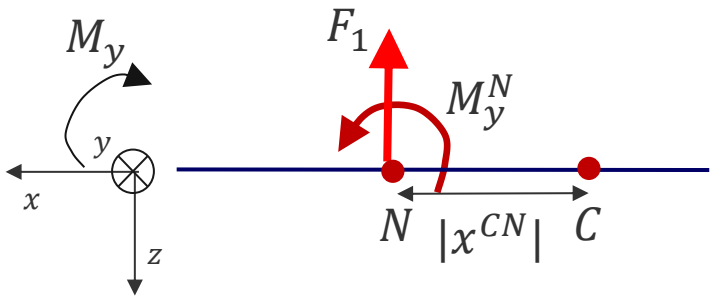




Analysis in the Plane:
Force acting at point C

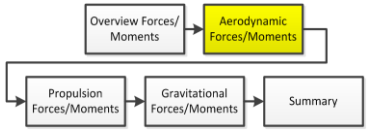
How can I transfer F_1 to point N without changing its effect on the body?

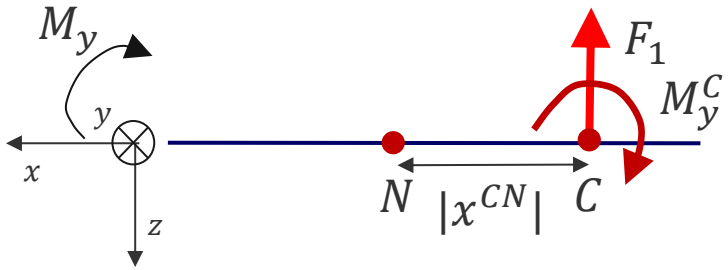
Add moment that F_1 produces around N



$$M_y^N = -F_1 \cdot |x^{CN}|$$

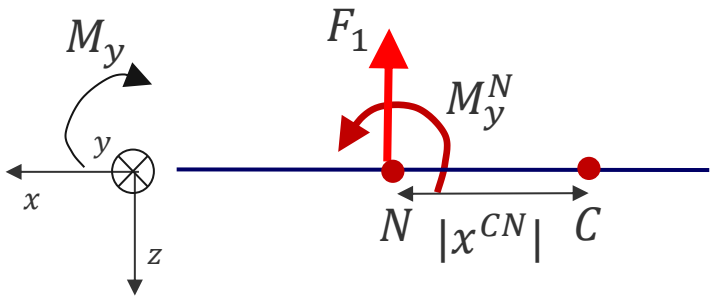
Note: keep track on signs!





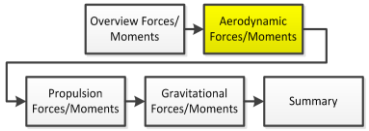
Analysis in the Plane:
Force and moment acting at point C

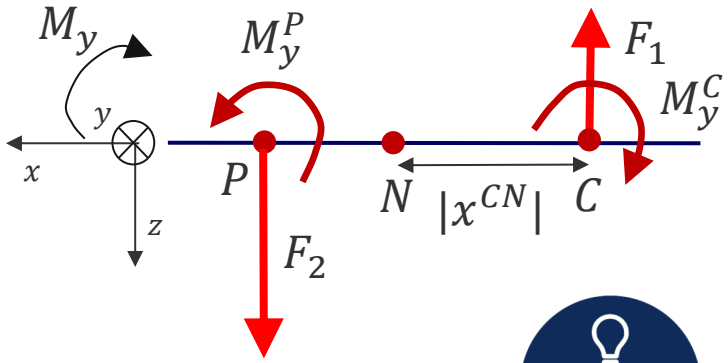
Transfer F_1 to point N without changing its effect on the body – What is the resulting moment with respect to point N?



$$M_y^N = -F_1 \cdot |x^{CN}| + \mathrm{M_y^C}$$

Note: keep track on signs!

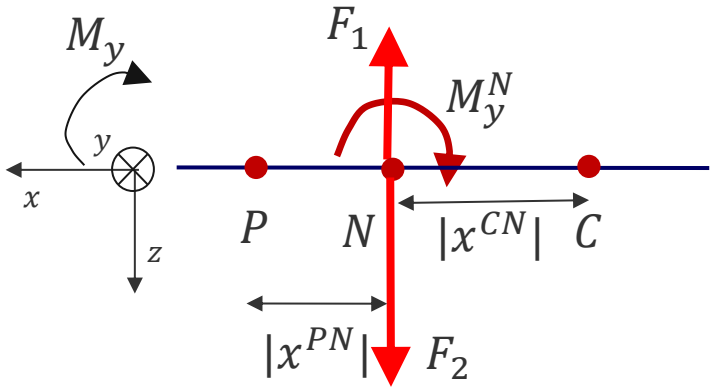




Analysis in the Plane:
Force and moment acting at point C

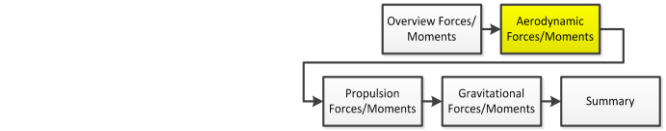


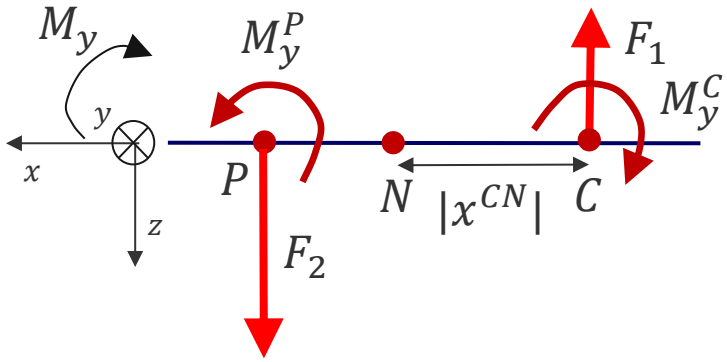
Transfer the forces to point N without changing its effect on the body – What is the resulting moment with respect to point N?



$$M_y^N = -F_1 \cdot |x^{CN}| + \mathrm{M}_y^C - F_2 \cdot |x^{PN}| - \mathrm{M}_y^P$$

Note: keep track on signs!





Analysis for 3 dimensional scenario:
Force and moment acting at point C

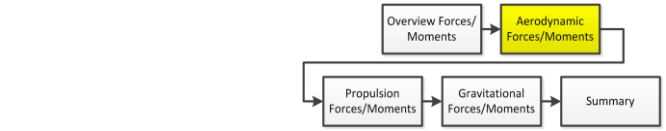
$$\boldsymbol{M}^N = \boldsymbol{x}^{NC} \times \boldsymbol{F}^C + \boldsymbol{M}^C + \boldsymbol{x}^{NP} \times \boldsymbol{F}^P + \boldsymbol{M}^P$$

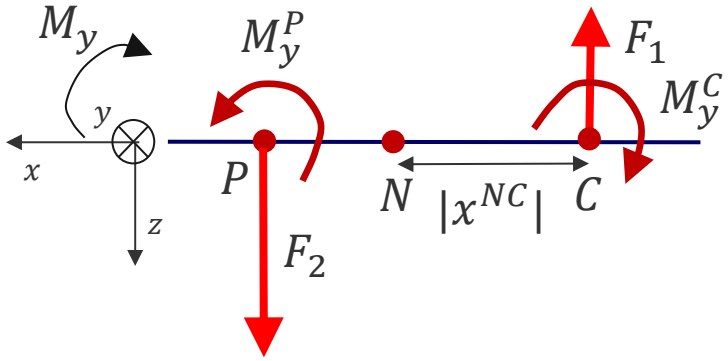
Note: with bold we denote vectors!
Make sure elements of vectors adhere to coordinate frame definition



Fill the gaps:

$$\boldsymbol{M}^N = \left(\quad\right) \times \left(\quad\right) + \left(\quad\right) + \left(\quad\right) \times \left(\quad\right) + \left(\quad\right)$$

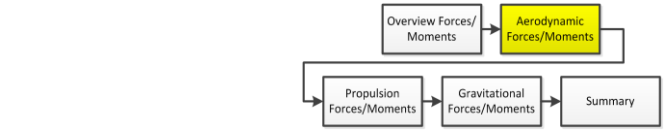


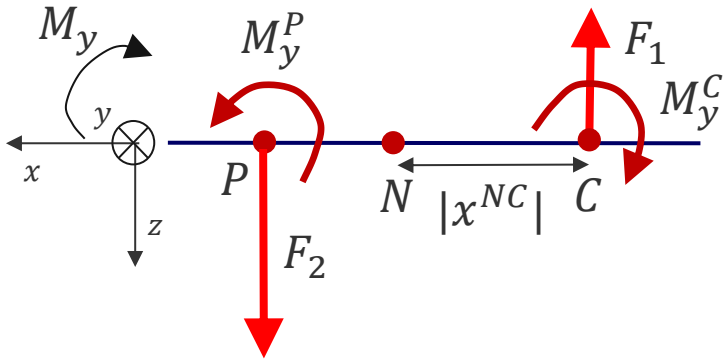


Analysis for 3 dimensional scenario:
Force and moment acting at point C

$$\boldsymbol{M}^N = \boldsymbol{x}^{NC} \times \boldsymbol{F}^C + \boldsymbol{M}^C + \boldsymbol{x}^{NP} \times \boldsymbol{F}^P + \boldsymbol{M}^P$$

$$\boldsymbol{M}^N = \begin{pmatrix} -|x^{NC}| \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F_1 \end{pmatrix} + \begin{pmatrix} 0 \\ M_y^C \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$





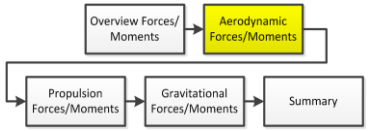
Analysis for 3 dimensional scenario:
Force and moment acting at point C

$$\boldsymbol{M}^N = \boldsymbol{x}^{NC} \times \boldsymbol{F}^C + \boldsymbol{M}^C + \boldsymbol{x}^{NP} \times \boldsymbol{F}^P + \boldsymbol{M}^P$$

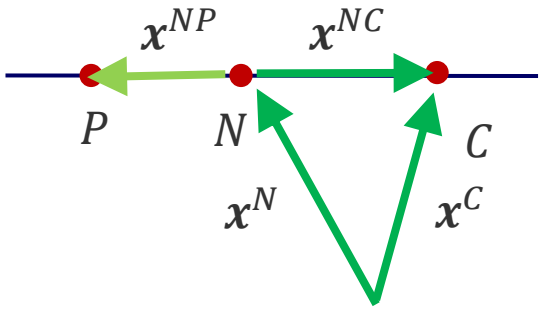
$$\boldsymbol{M}^N = \begin{pmatrix} -|x^{NC}| \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ -F_1 \end{pmatrix} + \begin{pmatrix} 0 \\ M_y^C \\ 0 \end{pmatrix} + \begin{pmatrix} |x^{NP}| \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ F_2 \end{pmatrix} + \begin{pmatrix} 0 \\ -M_y^P \\ 0 \end{pmatrix}$$

$$\boldsymbol{M}^N = \begin{pmatrix} 0 \\ -F_1 \cdot |x^{NC}| \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ M_y^C \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -F_2 \cdot |x^{NP}| \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -M_y^P \\ 0 \end{pmatrix}$$

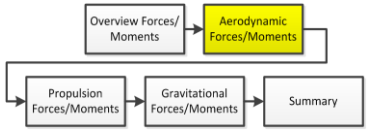
$$M_y^N = -F_1 \cdot |x^{NC}| + \mathrm{M_y^C} - F_2 \cdot |x^{NP}| - \mathrm{M_y^P}$$



Note:



$$x^N + x^{NC} = x^C \quad \Rightarrow \quad x^{NC} = x^C - x^N$$

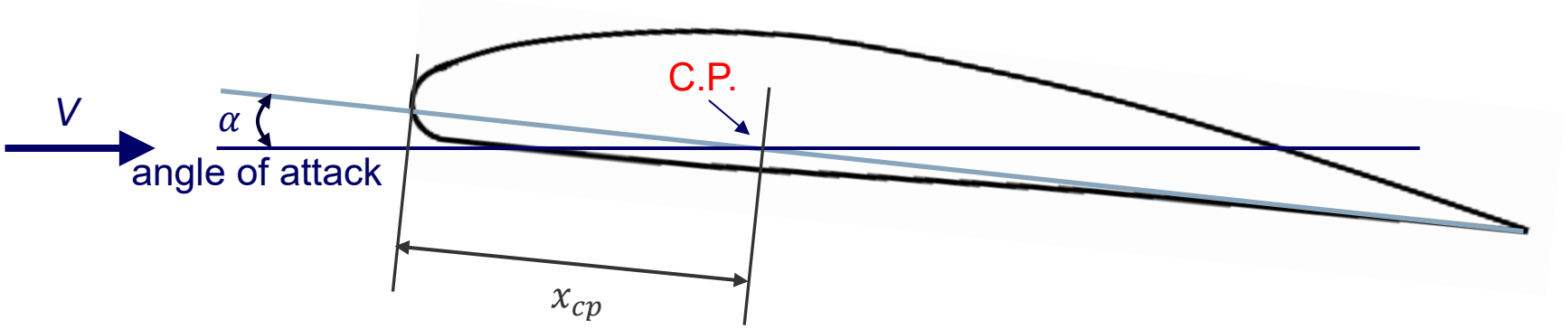


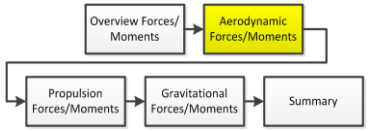


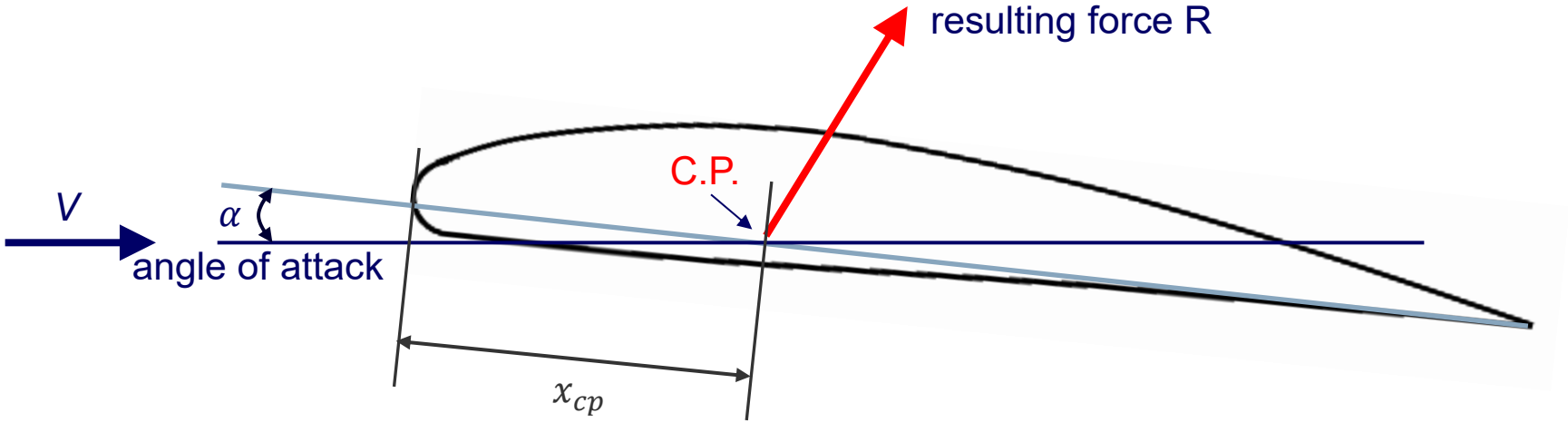
What are the characteristics of the center of pressure?

Hint: 3 key elements

- Force
- Moment
- Position





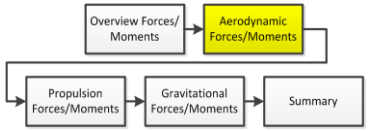


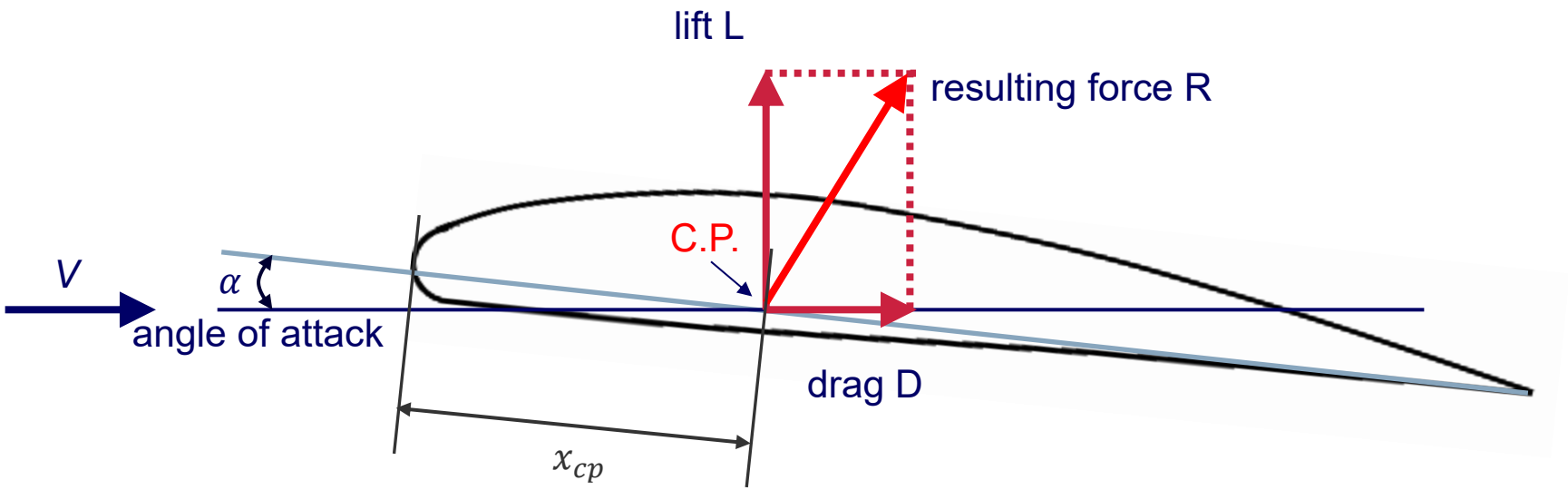
1.

- center of pressure (C.P.): point where total aerodynamic force acts (due to the pressure distribution around the profile)
- resulting aerodynamic force can be split into two components

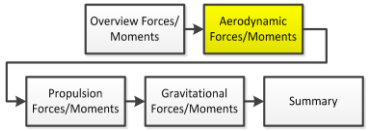


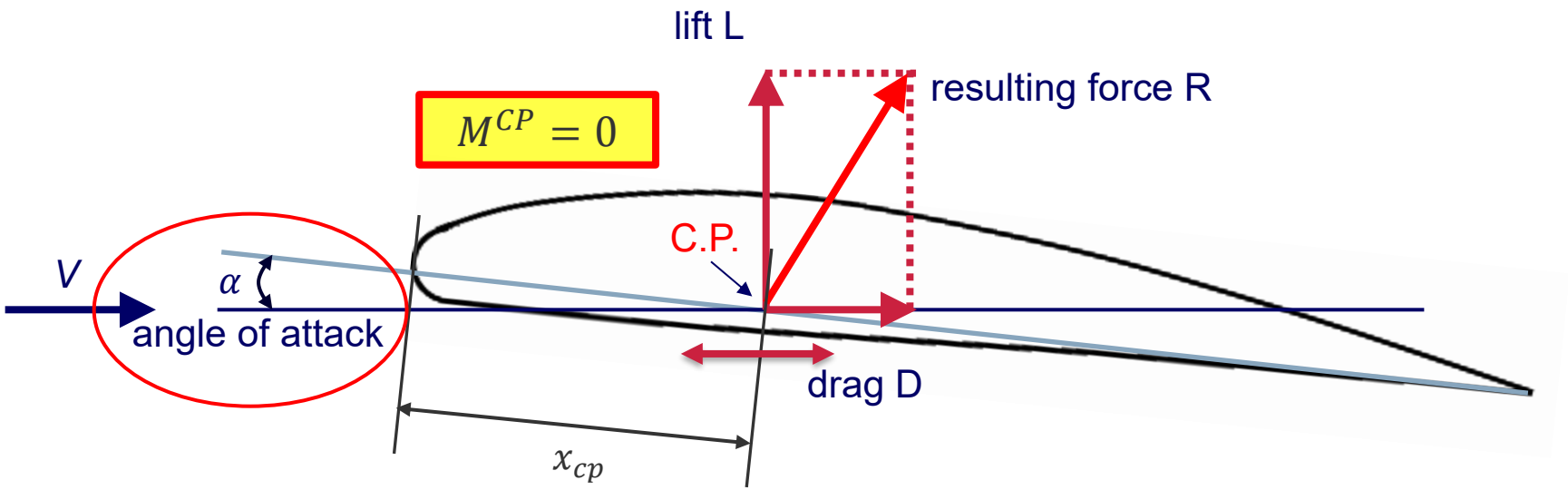
Can you draw and name the two components?





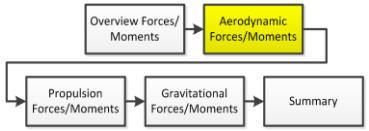
- center of pressure (C.P.): point where total aerodynamic force acts (due to the pressure distribution around the profile)
- resulting aerodynamic force can be split into two components
  - **Lift (*perpendicular to the direction of airflow (V)*)**
  - **Drag (*parallel to the direction of airflow (V)*)**





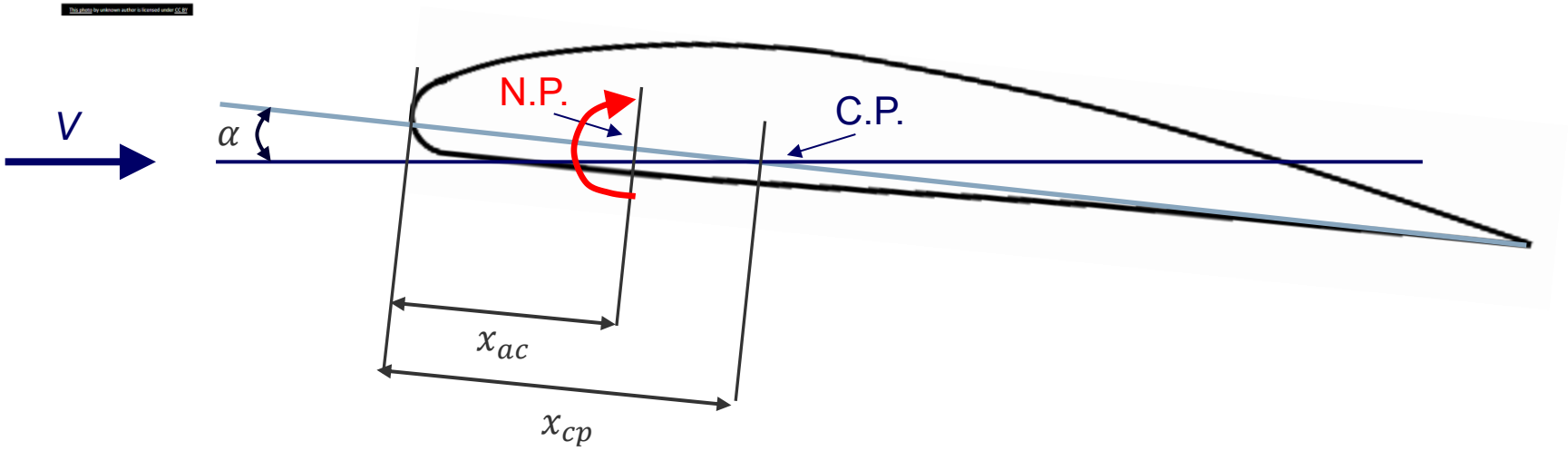
2. ▪ There is **no** total resulting moment around the center of pressure

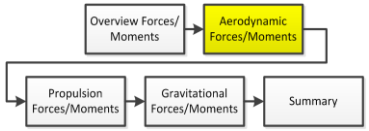
3. ▪ **C.P. position changes with angle of attack**

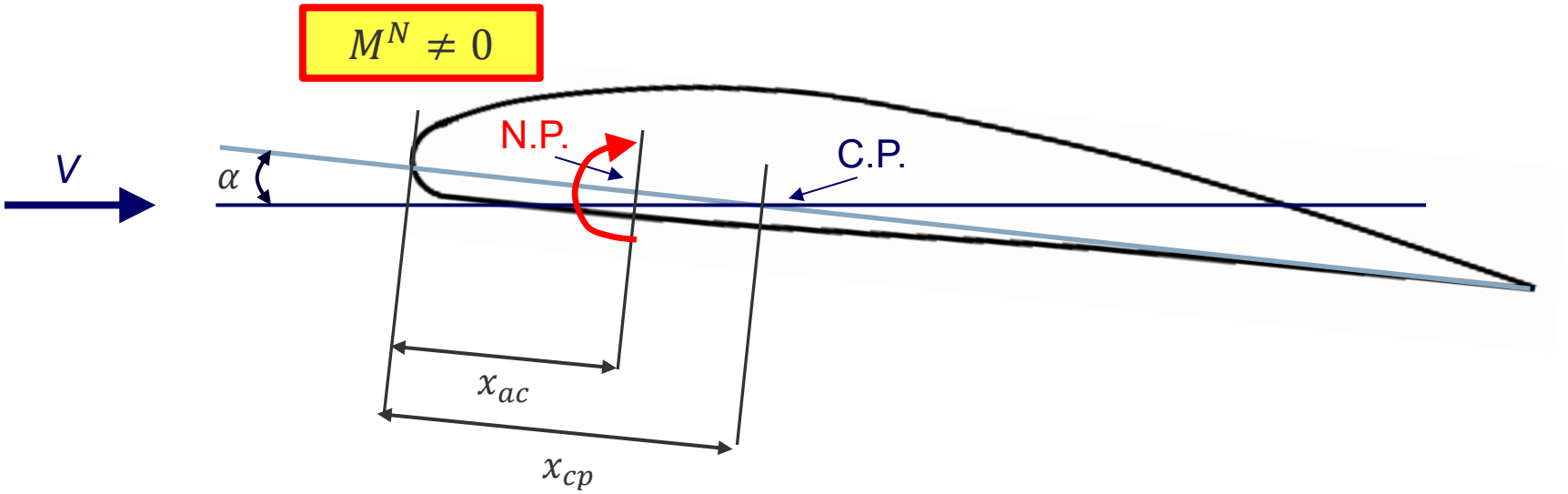




What are the characteristics of the neutral point? (aerodynamic point)







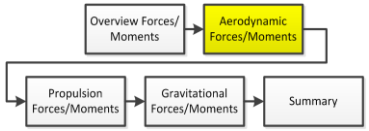
- neutral point: point where pitching moment does **not change** with the angle of attack

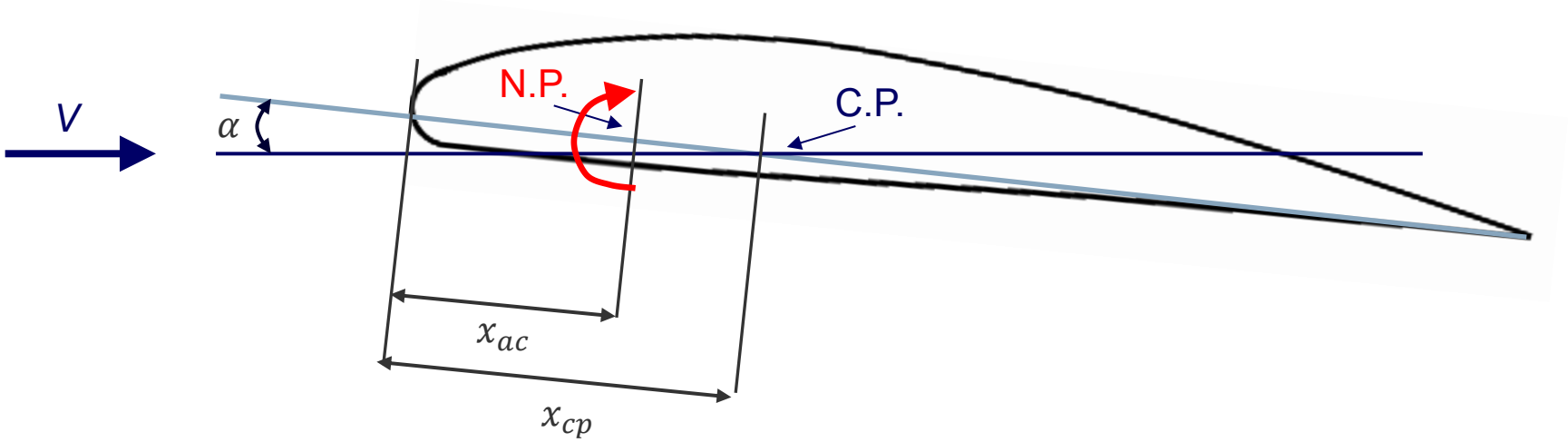
➢ The pitch moment does not depend on the angle of attack!

$$C_{M\alpha,NP} = \frac{\delta M^N}{\delta \alpha} = 0$$

➢ However, the pitch moment is not equal to 0 in the general case!

$$M^N \neq 0 \; ; C_{m,N} \neq 0$$



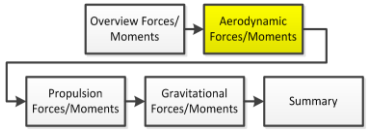


Position of N.P.:

subsonic flight: at around 25% of the cord line

supersonic flight: at around 50% of the cord line

Position is not changing with α

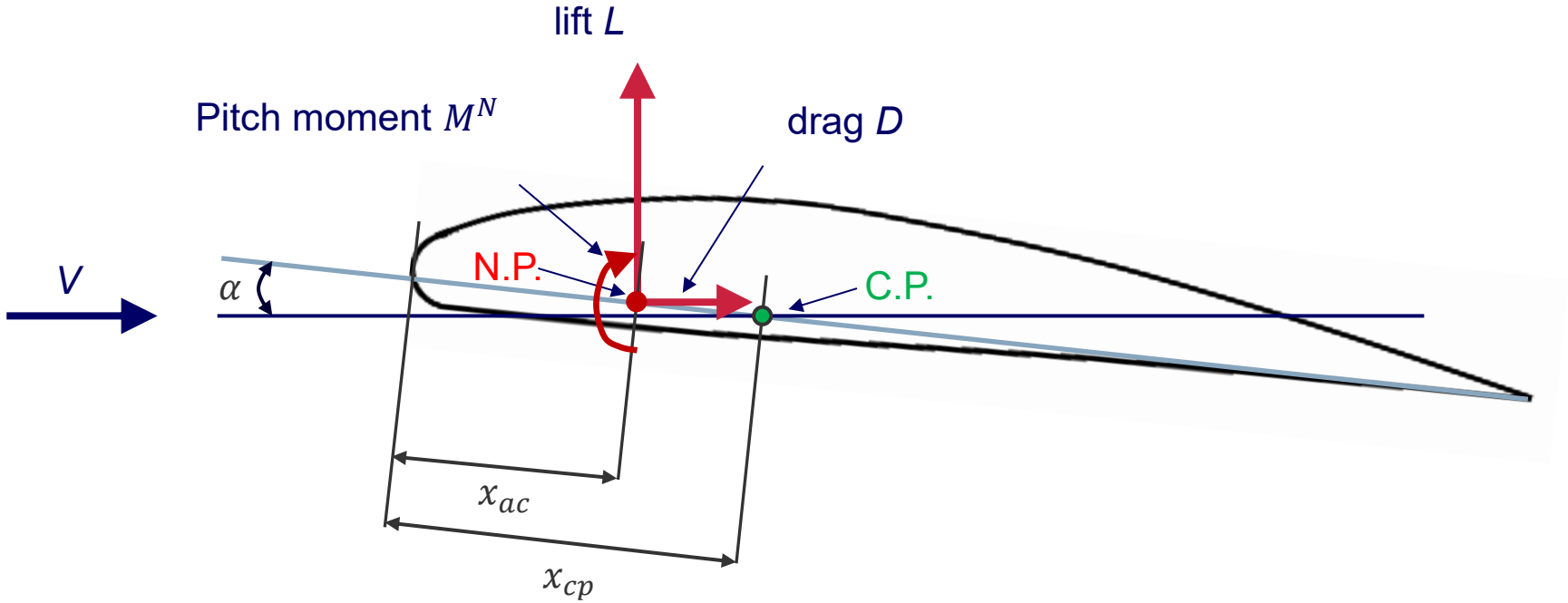


Exercise 1: Neutral Point of the Wing

a) What holds true for the pitch moment around the neutral point?

b) How does the position of the center of pressure depend on the angle of attack?

a) What statements hold true for the **pitch moment** around the neutral point N.P. and the center of pressure C.P. and their **positions**?

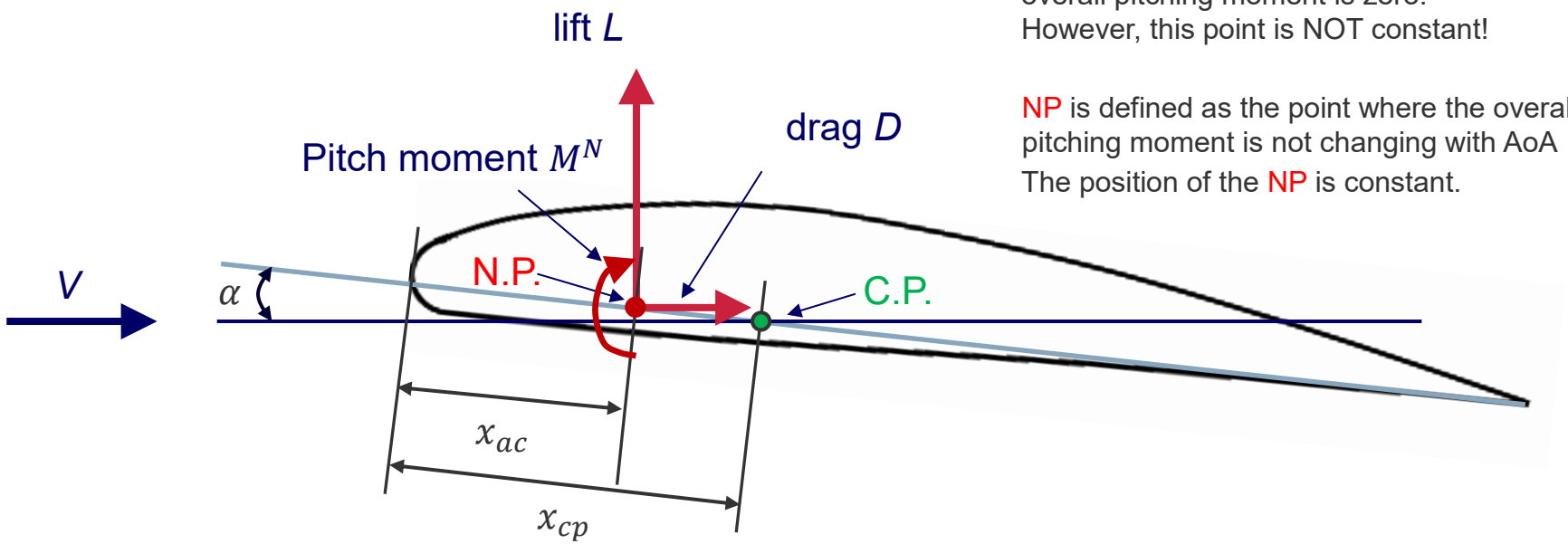


	Position	Pitching Moment
CP		
NP		

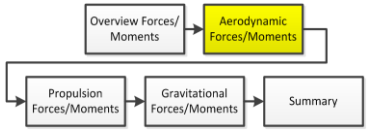
a) What statements hold true for the **pitch moment** around the neutral point N.P. and the center of pressure C.P.?

The CP is defined as the point where the overall pitching moment is zero.
However, this point is NOT constant!

NP is defined as the point where the overall pitching moment is not changing with AoA
The position of the NP is constant.



	Position	Pitching Moment
CP	$\frac{\delta x^{CP}}{\delta \alpha} \neq 0$, position changes with α	$M^{CP} = 0$
NP	$\frac{\delta x^{N}}{\delta \alpha} = 0$, position not changing with α	$M^{N} \neq 0$ but $\frac{\delta M^{N}}{\delta \alpha} = 0$, moment not changing with α



Exercise 1: Neutral Point of the Wing

a) What holds true for the pitch moment around the neutral point?

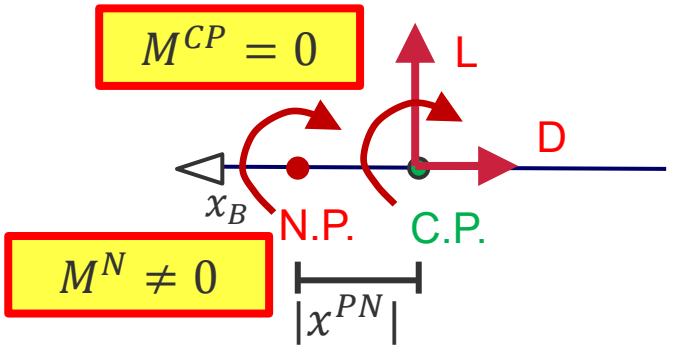
b) How does the position of the center of pressure depend on the angle of attack?

b) How does the position of the center of pressure depend on the angle of attack?

(Will the C.P. move forward (to the left) or backward if α is increased?)

Assumption:

You can neglect the Drag or assume that $z_{PN} = 0$



b) How does the position of the center of pressure depend on the angle of attack?

(Will the C.P. move forward (to the left) or backward if α is increased?)

Assumption:

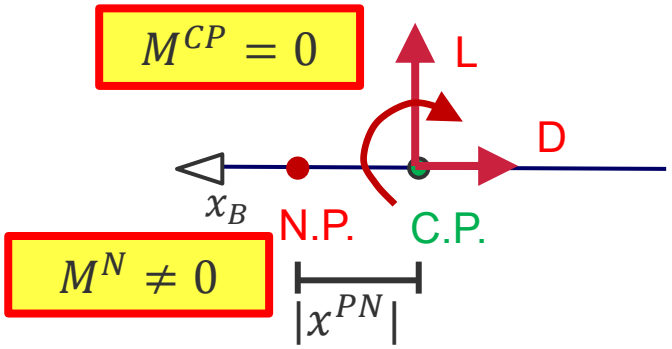
You can neglect the Drag or assume that $z_{PN} = 0$

I. What is the value of M^N?

$$M^{CP} = 0$$

$$M^N = M^{CP} - L \cdot |x^{PN}| = -L \cdot |x^{PN}|$$

$$-L \cdot |x^{PN}| = M^N = const.$$
(if AoA is changed)



II. What can we conclude from this relation regarding the α dependency of the C.P. position?

$$|x^{PN}| = -\frac{M^N}{L}$$

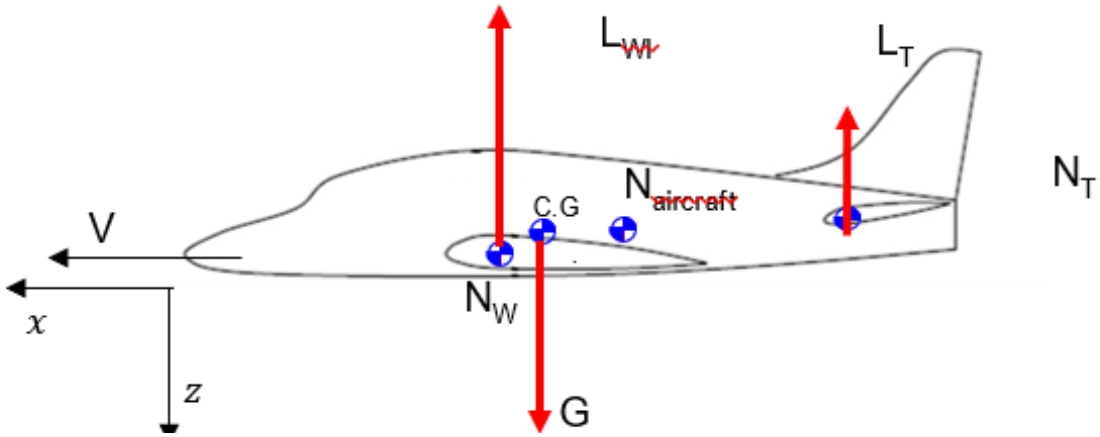
Because lift (usually) increases with AoA,…

… x^{PN} must decrease with increasing AoA!
Thus, the C.P. moves forward towards the N.P..

Exercise 2: Neutral Point of the Aircraft

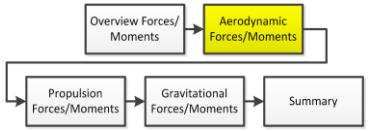
In this exercise we will investigate the neutral point N of the aircraft. We assume that,

- the aircraft is in horizontal steady state cruise flight without wind
- both the wing and horizontal stabilizer have a purely horizontal inflow
- drag and thrust can be neglected



a) Calculate the mass of the aircraft.
b) Calculate the position of the neutral point N of the aircraft.
c) Calculate the moment M^N in the neutral point.
Hint: Assemble forces and moments in the neutral point of the aircraft.

Component Build Up Method



Definitions for C.P. and N.P. are specified for **profiles**

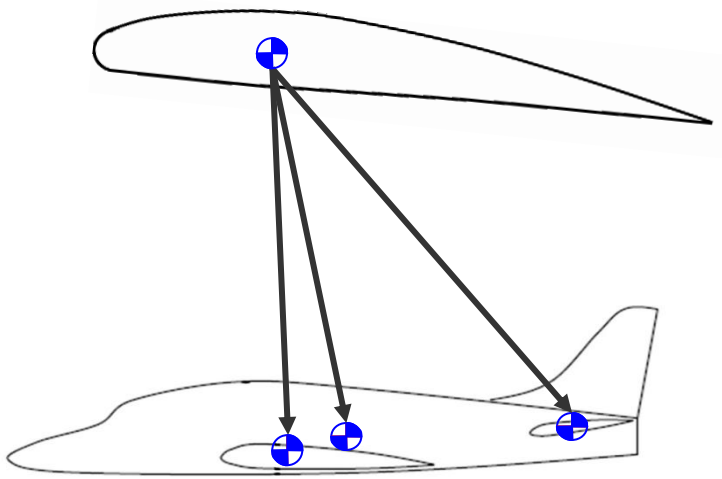
Aircraft have a more complex structure.

- not sufficient to calculate forces and moments only for the main wing.
- A more detailed approach is called **Component Build up method**

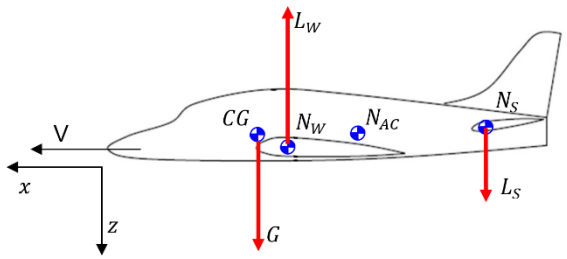
The approach calculates the **forces and moments for each part** of the aircraft **separately**:

1. wing
2. body
3. tail

The **total** forces and moments are calculated by **combining** the separate parts.



Neutral point of total aircraft configuration



Lift of the wing	$\|L_W\|$	17000 N
Change of the wing lift with the angle of attack	$\Delta L_{W,\alpha}$	2000 N/°
Position of the neutral point of the wing N_W	x_{N_W}	-2.3m
Lift of the stabilizer	$\|L_S\|$	1500 N
Change of the stabilizer lift with the angle of attack	$\Delta L_{S,\alpha}$	50 N/°
Position of the neutral point of the stabilizer N_S	x_{N_S}	-7m
Position C.G.	x_G	-1.8 m

1. The aircraft is in horizontal steady-state cruise flight without wind
2. Wing and horizontal stabilizer have a purely horizontal inflow
3. Drag and thrust are neglected

$g = 9.81 m/s^2$

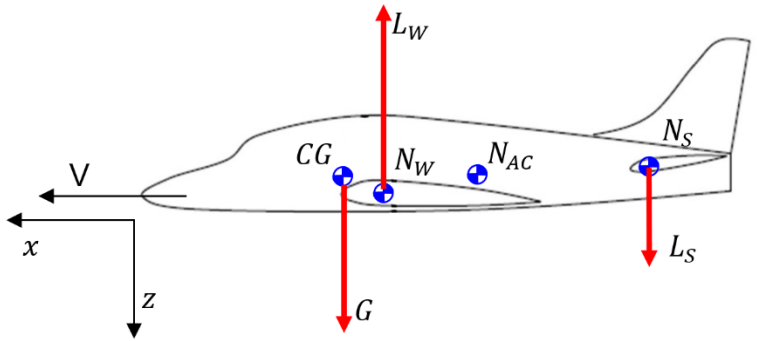
Neutral point of total aircraft configuration

a) Calculate the mass of the aircraft.

b) Calculate the position of the neutral point N of the aircraft.

c) Calculate the moment M^N in the neutral point.

Hint: Assemble forces and moments in the neutral point of the aircraft.



Neutral point of total aircraft configuration

a) Calculate the **mass of the aircraft**.

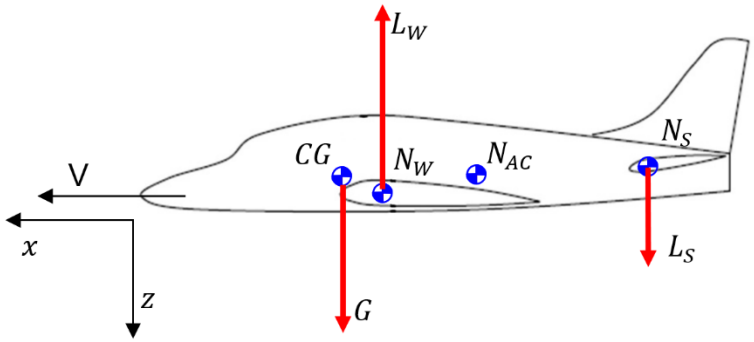
Recall assumption 1:
The aircraft is in horizontal **steady-state** cruise flight without wind

$$\sum F_i = 0 = F_G + F_W + F_S \quad \Rightarrow \quad mg - L_W + L_S = 0$$

We define the positive force direction downwards, hence: $F_W = -L_W$, $F_S = +L_S$)

$$\Rightarrow \quad m = \frac{L_W - L_S}{g} = 1580kg$$

Everything known here!



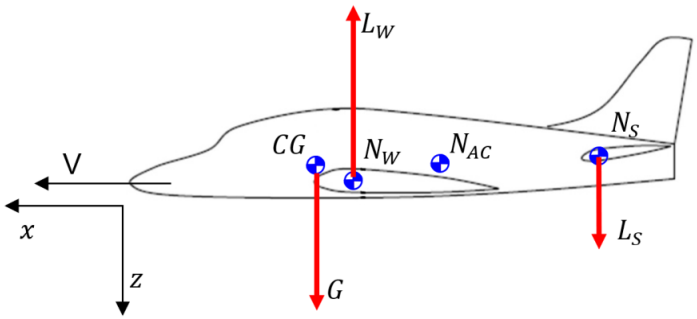
Neutral point of total aircraft configuration

a) Calculate the mass of the aircraft.

b) Calculate the position of the neutral point N of the aircraft.

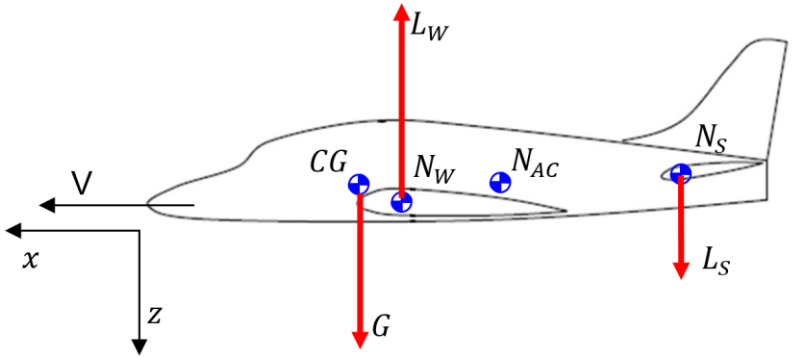
c) Calculate the moment M^N in the neutral point.

Hint: Assemble forces and moments in the neutral point of the aircraft.

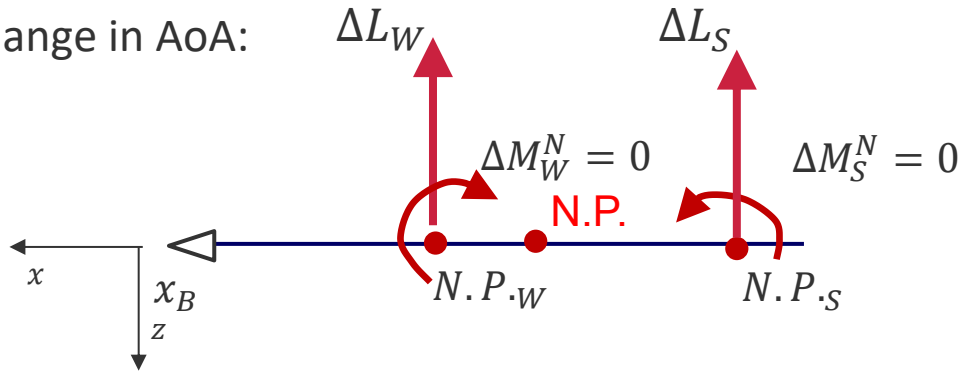


Neutral point of total aircraft configuration

b) Calculate the position of the neutral point of the aircraft.



Change in AoA:



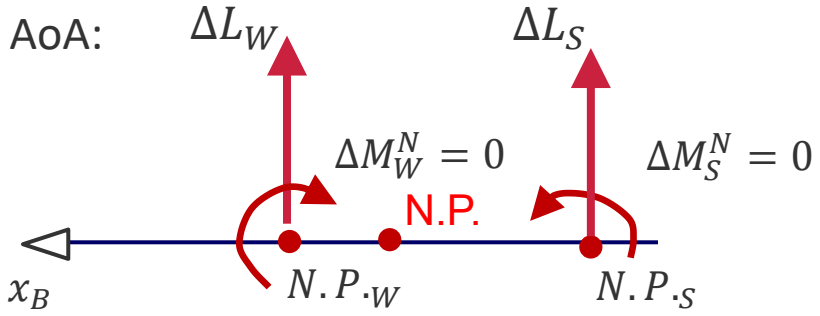
$F_G = mg$

does not change with α

Neutral point of total aircraft configuration

b) Calculate the position of the neutral point of the aircraft.

Change in AoA:



$F_G = mg$
does not change with α

The total pitching moment around the NP due to change of AoA:

$$\Delta M^N = \Delta M_W^N + \Delta L_W |x^{NN_W}| + \Delta M_S^N - \Delta L_S |x^{NN_S}|$$

we use $\Delta M_W^N = \Delta M_S^N = 0,$

$$\Delta M^N = \Delta L_W |x^{NN_w}| - \Delta L_S |x^{NN_s}|$$

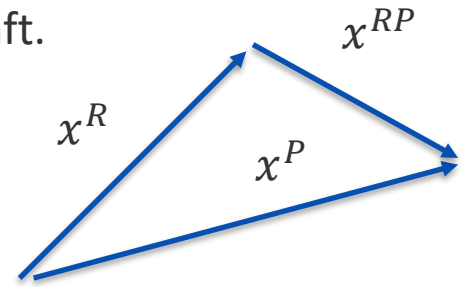
$x^{NN_w} > 0, x^{NN_s} < 0$

$$\Delta M^N = \Delta L_W x^{NN_w} + \Delta L_S x^{NN_s}$$

Neutral point of total aircraft configuration

b) Calculate the position of the neutral point of the aircraft.

$$\Delta M^N = \Delta L_W x^{NN_w} + \Delta L_S x^{NN_s}$$



$$\Delta M^N = \Delta L_W (x^{N_w} - x^N) + \Delta L_S (x^{N_s} - x^N)$$

$$x^R + x^{RP} = x^P$$

$$x^{RP} = x^P - x^R$$

$$\Delta M^N = -\Delta L_W x^N + \Delta L_W x^{N_w} - \Delta L_S x^N + \Delta L_S x^{N_s} = 0$$

The total pitching moment around the NP will not vary with AoA! Hence the change of M^N with α is zero, i.e. $\Delta M^N = 0$

$$x^N (\Delta L_W + \Delta L_S) = \Delta L_W x^{N_W} + \Delta L_S x^{N_S}$$

$$x^N = \frac{\Delta L_W x^{N_W} + \Delta L_S x^{N_S}}{\Delta L_W + \Delta L_S} = -2.41\ m$$

Everything known here!

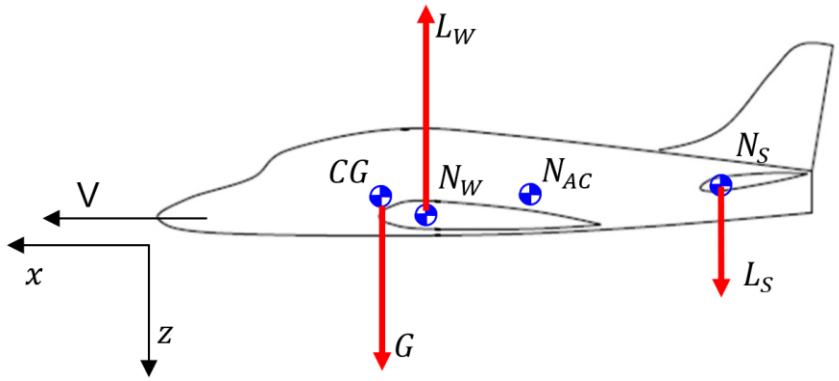
Neutral point of total aircraft configuration

a) Calculate the mass of the aircraft.

b) Calculate the position of the neutral point N of the aircraft.

c) Calculate the moment M^N in the neutral point.

Hint: Assemble forces and moments in the neutral point of the aircraft.

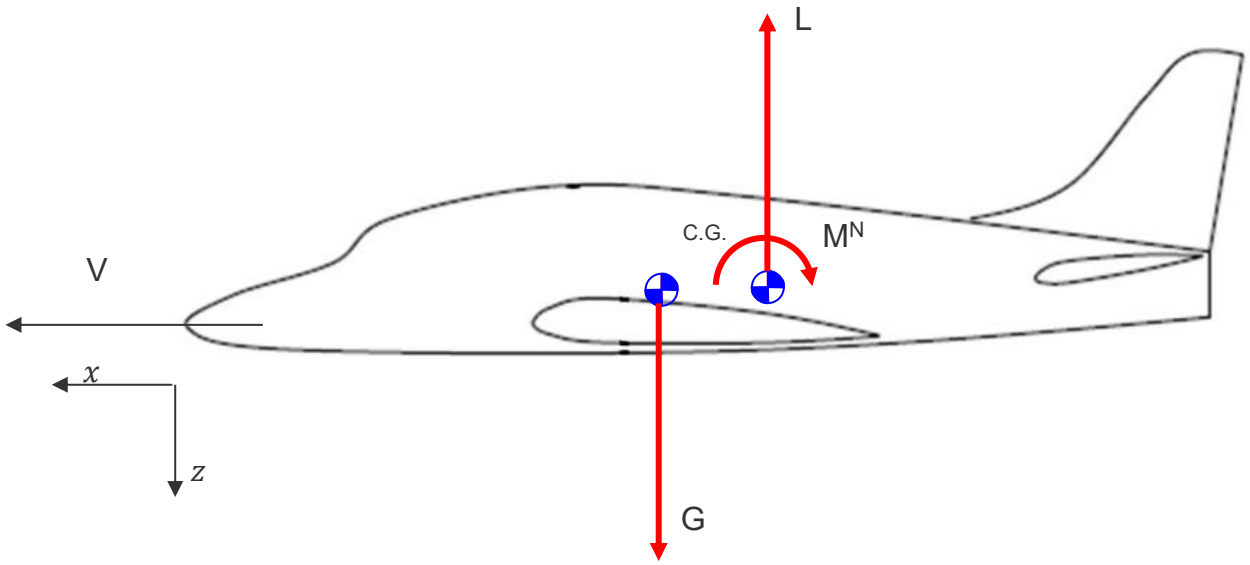


Neutral point of total aircraft configuration

c) Calculate the **moment in the neutral point**.

Recall assumption 1:

The aircraft is in horizontal steady-state cruise flight without wind



Neutral point of total aircraft configuration

c) Calculate the **moment in the neutral point**.

Recall assumption 1:

The aircraft is in horizontal **steady-state** cruise flight without wind

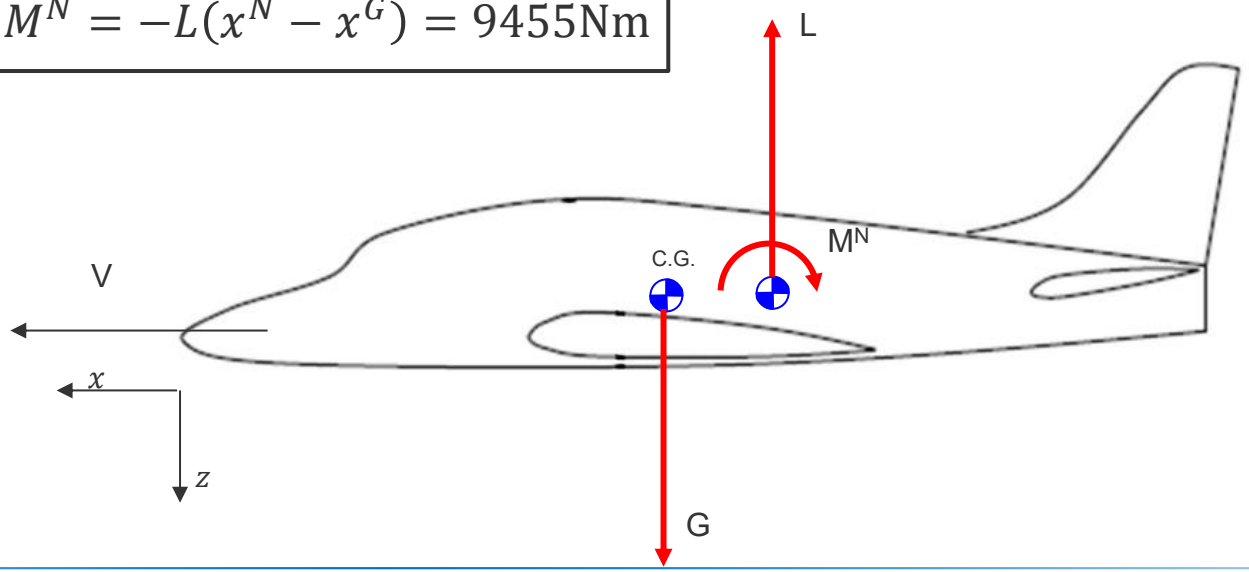
$$\sum M^G = 0, \sum F = 0$$

$$L = G = \mathrm{L_W} + \mathrm{L_S} = 15500\mathrm{N}$$

$$x^{GN} < 0$$

$$\sum M^G = -L \cdot |x^{GN}| + M^N = L(x^N - x^G) + M^N = 0$$

$$\boxed{M^N = -L(x^N - x^G) = 9455\mathrm{Nm}}$$





Introduction To Flight Mechanics & Control

Exercise 3